Applications

1

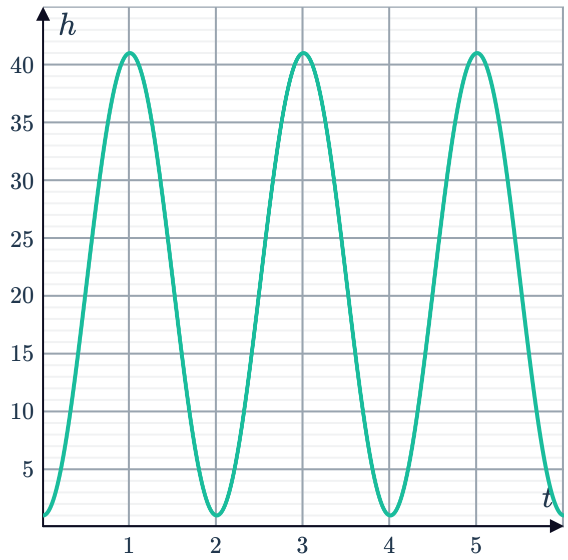
- **a** Sine, the graph's initial value is at the midline.
- **b** $I = \sin(90t)$
- **c** 210 times

2

- **a** 6 seconds
- **b** 2500 mL
- **c** 200 mL
- **d** 60° per second
- **e** $c = \frac{1}{2}$

3

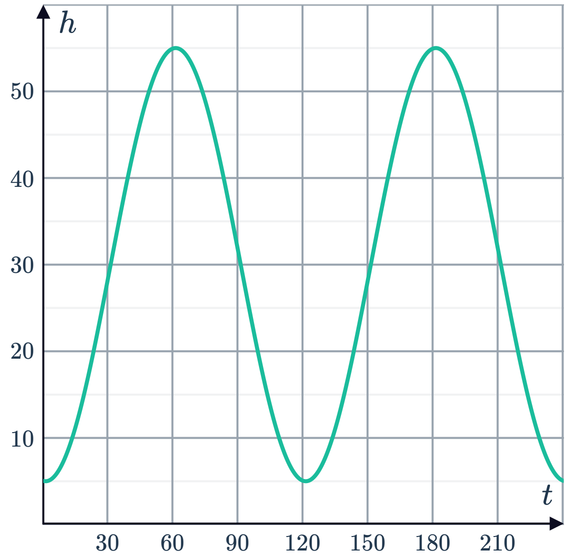
- **a**
  - **i** No
  - **ii** No
  - **iii** No
  - **iv** Yes
- **b**
  - **i** 2 minutes
  - **ii** 1 m
  - **iii** 41 m
- **c**



- **d** $h = -20\cos(180t) + 21$
- **e** $t = \frac{1}{2}$ minute
- **f** The height of the passengers above or below the centre of the wheel at time t.

4

a



b 55 m

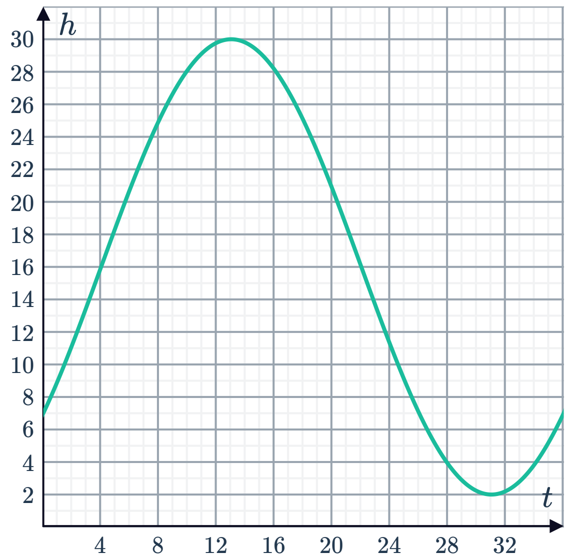
c 5 m

d 36.47 m

e 120 seconds

5

a



b 7.00 m

c $t = 34.82$ seconds

d 3

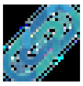
e 3.88 m

6

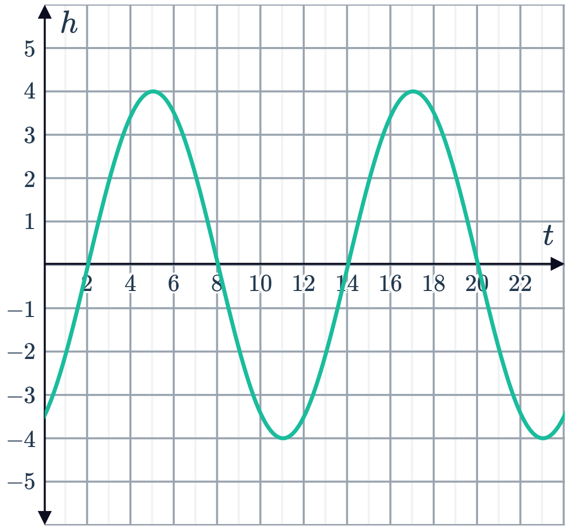
a -2 m

b 5:00 am

c 6 hours



d



e 0 m

f 0.14 m

7

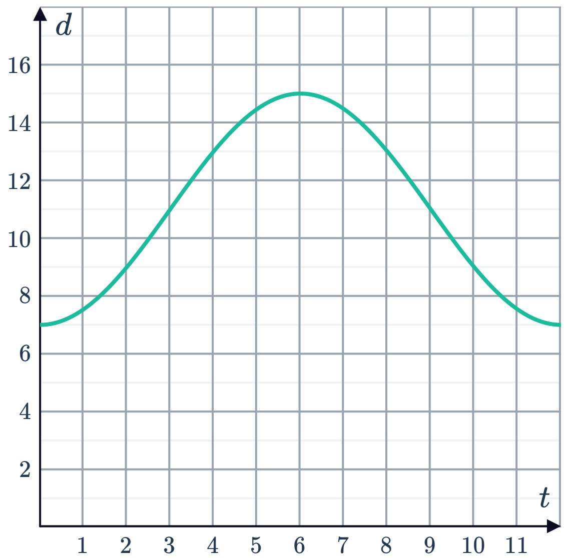
a

Time (t, hours)	0	3	6	9	12
Tide level (d, metres)	7	11	15	11	7

b 4 m

c 12 hours

d



e

i Yes **ii** No **iii** No **iv** No

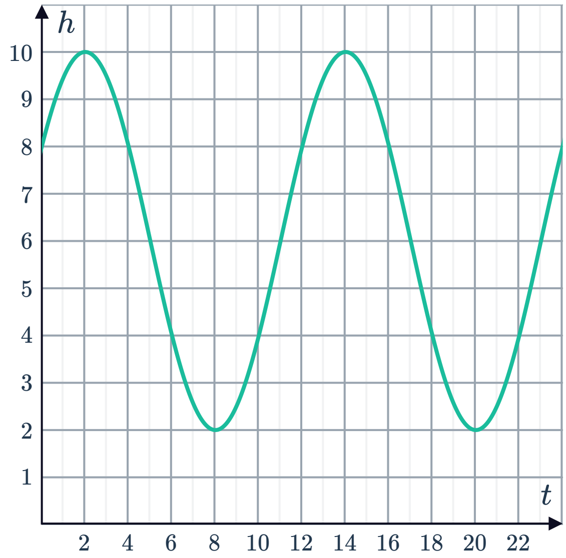
f $d = -4\cos 30t + 11$

g 3.34 hours

8

a $h(t) = 6 + 4\cos(30(t-2))$

b



c 2:00 am

d 8:00 am

9

a

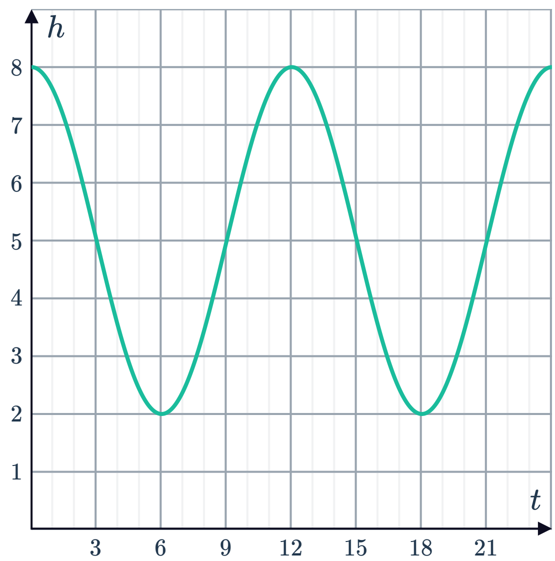
i $d = 5\text{ m}$

ii $a = 3\text{ m}$

iii $b = 30^{\circ}$ per hour

iv $c = 90^{\circ}$

b



c 6 and 18 hours after noon

10

a

i $a = 40\text{ cm}$

ii $d = 10\text{ cm}$

iii $c = \dfrac{1}{4}$

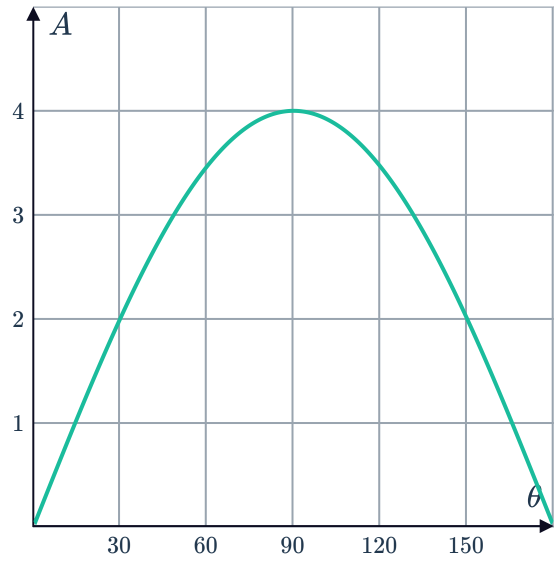
b $t = \dfrac{1}{3}$ second

11

a $A = 4\sin\theta$

b $0 < \theta < 180$

c



d 30°

12

a 5 cm to the right of the centre

b To the left

c 1 second

d 4 seconds

e $x = 5\cos(90t)$